M1: CBD-Net

Implementation and results report

Oil-spill detection in Sentinel-1 SAR imagery · method 1 of 3

Method. Q. Zhu et al., “Oil spill contextual and boundary-supervised detection network based on marine SAR images,” IEEE TGRS [P24]. Official code: github.com/CUG-URS/CBDNet-main.

What this report contains. Two trained models of the same method and four tests: (A) the official SOS test split, which is **leaky**; (B) a **leak-free** group-disjoint split; (C) the external Peru dataset; and (D) the external MDOD dataset. Every table is generated from the saved metric files, and the terminal figures are excerpts of the real console logs.

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1. Summary of results

Table 1. All M1 results at a glance (%, test-time augmentation unless noted).

Test	Model	Oil F1	Oil IoU	mIoU	Precision	Recall	ECE (×100)
A. SOS official split (leaky)	official-split	87.91	78.43	82.79	84.43	91.69	2.83
published [P29]		87.81	—	82.94	—	—	—
B. SOS group split (leak-free)	leak-free	86.80	76.68	83.06	86.59	87.01	2.64
C. Peru (external, 2,013 patches)	official-split	24.46	13.94	37.16	14.00	96.82	26.47
	leak-free	29.44	17.26	43.51	17.41	95.29	18.62

Table 2. MDOD (external, box labels only): share of boxes flagged as oil, and how well oil is ranked above look-alikes.

Model	Oil boxes flagged (should be high)	Look-alike boxes flagged (should be low)	AUC oil vs look-alikes (0.5 = chance)
official-split	64.0	91.8–99.8	0.282
leak-free	55.5	87.0–98.2	0.335

Key findings

- **The reproduction succeeds.** On the official SOS split our mIoU is 82.79 % (published 82.94 %) and oil F1 is 87.91 % (published 87.81 %); both gaps are within ±0.2 pp.
- **The official split leaks, but the effect is small.** Every official test tile has augmented copies of its sea area in training. On a leak-free split, oil IoU drops by 1.74 pp and oil F1 by 1.11 pp.
- **M1 does not transfer to a new sea.** On Peru, oil F1 falls to 24.5–29.4 %. Recall stays above 95 %, but precision collapses to 14.0–17.4 % because almost every dark area is called oil. The probabilities become badly calibrated (ECE ×7–9).
- **M1 cannot reject look-alikes.** On MDOD every look-alike class is flagged more often than real oil, and the ranking AUC is below 0.5.
- **The leak-free model is the better model** on both external datasets (higher F1 and lower ECE on Peru, fewer false alerts on MDOD). All results are from a single training seed.

2. Method and implementation

2.1 How CBD-Net works

- **Input:** one 256 × 256 SAR tile (3 channels).
- **Encoder:** ResNet-34 pretrained on ImageNet, with an scSE attention block after every stage. scSE re-weights both the channels and the pixel locations.
- **Bottleneck:** a dilated-convolution block (rates 1, 2, 4, 8) that gives wide spatial context.
- **Decoder:** LinkNet-style, four up-sampling blocks with additive skip connections.
- **Output:** a sigmoid gives an oil probability for every pixel; threshold 0.5 gives the oil mask.

2.2 Training and testing setup

Table 3. Setup used for both models (identical to the official code).

Setting	Value
Training data	SOS Sentinel-1 training tiles, 256 × 256 (3,354 official / 3,358 leak-free)
Loss	BCE + Dice (as in the released code)
Optimiser	Adam, learning rate 2×10^{-4}, batch size 4
Schedule	Up to 80 epochs. After more than 3 epochs without improvement of the training loss: reload the best weights and divide the learning rate by 5. Stop after more than 6.
Augmentation	HSV jitter, shift-scale-rotate, horizontal / vertical flips, 90° rotations (official)
Testing	Official 8-way test-time augmentation (TTA): 4 rotations × flip, probabilities averaged. A single-pass result is also reported.
Parameters	31.4 M
Hardware	NVIDIA RTX 3050 laptop GPU (4 GB), 16 GB RAM, PyTorch 2.6 (CUDA 12.4)
Training time	official split: 71 epochs, 1 h 47 min; leak-free split: 50 epochs, 1 h 18 min

2.3 What we changed, and two things we found in the official code

The network, loss and schedule are imported **unchanged** from the official repository. We changed only:

- `np.math` → `math`, required by NumPy 2 (identical behaviour);
- `shuffle=True` in the data loader (the official code uses `shuffle=False`);
- fixed random seeds and a per-epoch CSV log, so that runs can be repeated.

Findings in the official code:

1. **The official evaluation script reports precision, recall and F1 for the background class, not for oil.** `t1p3-Iou.py` calls `confusion_matrix(labels=[255, 0])`, which makes index 1 the background. We report oil-class metrics as the main numbers and the official-script value separately.
2. **The released code trains with BCE + Dice only.** The boundary loss described in the paper exists in the repository (`jointloss`) but is not used by default.

2.4 Code map

Table 4. Where each part of M1 lives (repository root).

Part	File	Written by
Model (CBD-Net)	`third_party/CBDNet-main/networks/CBDNet.py`	official authors, unchanged
Loss (BCE + Dice)	`third_party/CBDNet-main/dice_bce_loss.py`	official authors, unchanged
Training loop	`M1_CBDNet/train_m1.py`	us, following the official `train.py`
Data loading, augmentation	`common/sos_data.py`	us; augmentation copied from the official code
SOS testing and metrics	`common/evaluate.py`	us; shared by M1–M3
Peru test	`common/evaluate_peru.py`	us; shared by M1–M3

Part	File	Written by
MDOD test	`common/evaluate_mdod.py`	us; shared by M1–M3
Reports and figures	`M1_CBDNet/make_report.py`, `make_peru_report.py`, `make_mdod_report.py`	us

3. Metrics

Pixels are pooled over all test tiles, as in the official SOS script. “Oil” is the positive class.

Table 5. Metric definitions (TP / FP / FN / TN = true / false positive / false negative / true negative oil pixels).

Metric	Meaning	Formula
Accuracy	share of all pixels classified correctly	(TP + TN) / all
Precision (oil)	of the pixels called oil, the share that really is oil	TP / (TP + FP)
Recall (oil)	of the real oil pixels, the share that was found	TP / (TP + FN)
F1 (oil)	balance of precision and recall	2PR / (P + R)
IoU (oil)	overlap of predicted and true oil areas	TP / (TP + FP + FN)
mIoU	mean of oil IoU and background IoU	(IoU_oil + IoU_bg) / 2
Boundary F1	F1 of slick outlines, within 2 pixels	—
ROC-AUC / PR-AUC	how well probabilities rank oil above non-oil (threshold-free)	area under curve
ECE	calibration: gap between predicted probability and observed frequency (15 bins); 0 = perfect	$\Sigma\ (n_b/N)\cdot\lvert acc_b - conf_b\rvert$
Brier score	mean squared error of the probabilities	mean $(p - y)^2$

4. Test A: SOS official split (leaky)

Run `M1_cbdnet_sentinel_dicebce_s0`. Trained on the 3,354 official Sentinel-1 training tiles and tested on the 839 official test tiles. Training stopped at epoch 71; the best training loss was 0.3515 at epoch 66.

4.1 Quantitative results

Table 6. Segmentation metrics on the official SOS Sentinel-1 test set.

Metric (%)	Ours, with TTA	Ours, single pass	Published [P29]
Accuracy	91.24	91.08	—
Precision (oil)	84.43	84.14	—
Recall (oil)	91.69	91.57	—
F1 (oil)	87.91	87.70	87.81
IoU (oil)	78.43	78.09	—
IoU (background)	87.15	86.92	—
mIoU	82.79	82.51	82.94
Boundary F1 (2 px)	37.82	37.24	—
F1 as in the official script (background class)	93.14	93.00	—

Table 7. Probability quality.

Metric	With TTA	Single pass
ROC-AUC	0.9748	0.9737
PR-AUC	0.9585	0.9569
ECE (15 bins)	0.0283	0.0329
Brier score	0.0639	0.0657

Reproduction gap: mIoU -0.15 pp and F1 +0.10 pp (target: within ±2 pp). TTA adds about 0.2–0.3 pp to every metric.

4.2 Training

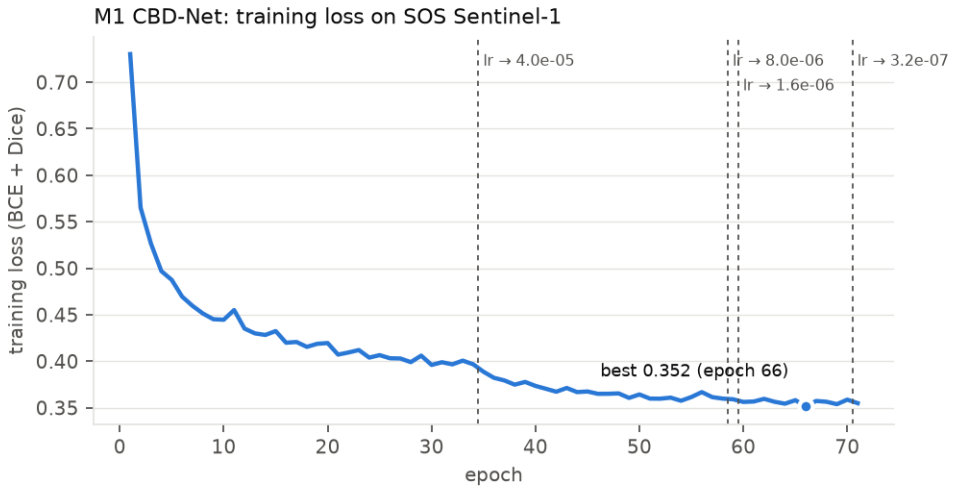


Figure 1. Training loss per epoch, official split. Dashed lines mark the learning-rate drops.

```
train_eval_sentinel_official.log  (official split)

M1_cbdnet_sentinel_dicebce_s0: 3354 train tiles, 839 iters/epoch
epoch   1  loss 0.7303  lr 2.0e-04  95s  *
epoch   2  loss 0.5651  lr 2.0e-04  180s  *
epoch   3  loss 0.5264  lr 2.0e-04  265s  *
epoch   4  loss 0.4969  lr 2.0e-04  350s  *
epoch   5  loss 0.4874  lr 2.0e-04  436s  *
...
  lr -> 4.0e-05 (reloaded best)
...
  lr -> 8.0e-06 (reloaded best)
...
  lr -> 1.6e-06 (reloaded best)
...
epoch  67  loss 0.3573  lr 1.6e-06  6060s
epoch  68  loss 0.3565  lr 1.6e-06  6152s
epoch  69  loss 0.3539  lr 1.6e-06  6245s
epoch  70  loss 0.3587  lr 1.6e-06  6340s
  lr -> 3.2e-07 (reloaded best)
epoch  71  loss 0.3549  lr 3.2e-07  6430s
done M1_cbdnet_sentinel_dicebce_s0 best train loss 0.35153
[tta] acc 0.9124  P 0.8443  R 0.9169  F1 0.8791  IoU_oil 0.7843  mIoU 0.8279  F1_bg(official) 0.9314  BF1 0.3782  AUC 0.9748  ECE 0.0283  Brier 0.0639
[plain] acc 0.9108  P 0.8414  R 0.9157  F1 0.8770  IoU_oil 0.7809  mIoU 0.8251  F1_bg(official) 0.9300  BF1 0.3724  AUC 0.9737  ECE 0.0329  Brier 0.0657
```

Figure 2. Console output of the official-split run (excerpt of `train_eval_sentinel_official.log`; "…" marks omitted epochs, * marks a new best loss). The last two lines are the test results.

4.3 Qualitative results

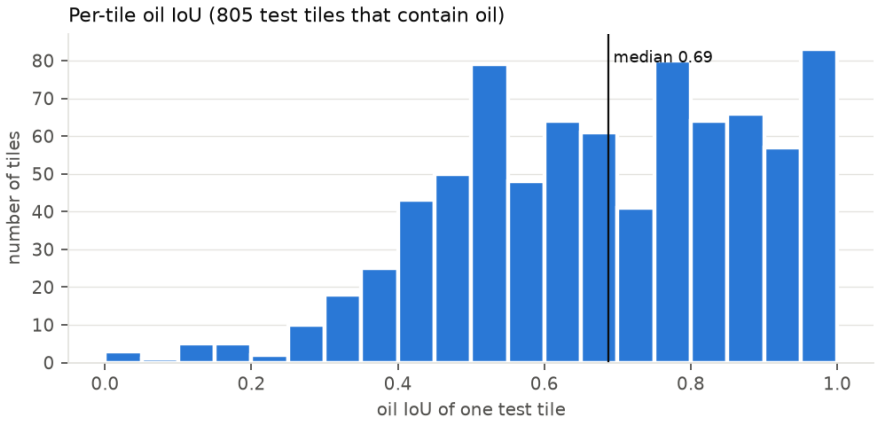


Figure 3. Per-tile oil IoU on the official test set. Median 0.69; 3 % of oil tiles score below 0.3.

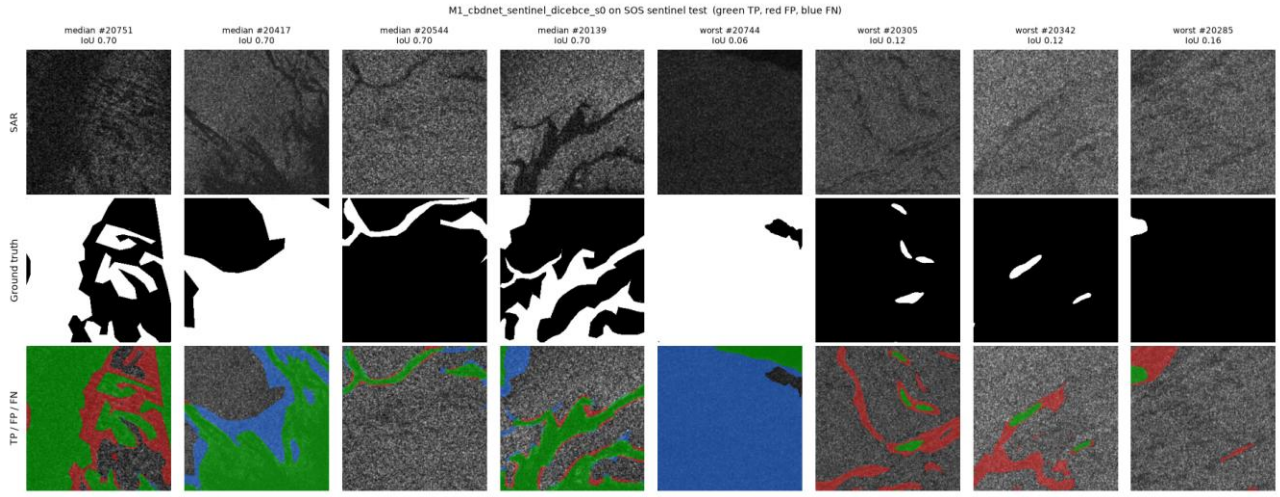


Figure 4. Four median and four worst test tiles. Green = correct oil, red = false oil, blue = missed oil.

- Large slicks are segmented well. Most errors sit on slick edges, where the ground-truth outlines are coarse polygons; this explains the low boundary F1 (37.82 %).
- The worst tiles are (a) fully dark tiles labelled as oil, which show no edge or context, and (b) thin dark streaks that the model marks as oil but the labels do not (possible look-alikes or missing labels).

- In-domain, the probabilities are already well calibrated (ECE 0.028).

5. Why the official split is leaky

SOS tile names are plain numbers with no scene information. Our near-duplicate check (template matching over all rotations and flips) found that the Sentinel-1 tiles 20001–22220 are 5 augmented versions of the same 444 sea areas: tile i and tile i + 444·k show the same area. The official test set is simply the first 839 IDs, so every official test tile has at least two augmented copies of its sea area in the training set.

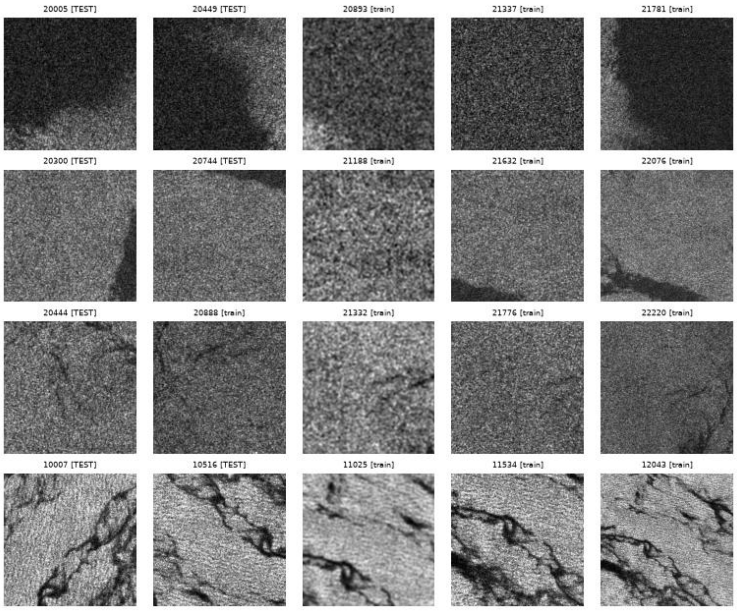


Figure 5. Examples of one sea area appearing as five different SOS tiles (IDs i, i+444, i+888, …).

Test A therefore shows that the **reproduction is correct**, but it does not measure performance on unseen sea areas. Test B removes this leak.

6. Test B: SOS group-disjoint split (leak-free)

Run `M1_cbdnet_sentinel_dicebce_s0_groupsplit`. Same model, loss, schedule and seed as Test A; only the split changes. Tiles are grouped by sea area, and whole groups are assigned to train or test, so **no sea area appears in both**. Train: 3,358 tiles (about the official 3,354, so training size is not a factor). Test: 835 tiles. Training stopped at epoch 50; best training loss 0.3670 at epoch 46.

6.1 Quantitative results

Table 8. All metrics on the leak-free test set.

Metric (%)	With TTA	Single pass
Accuracy	92.16	91.91
Precision (oil)	86.59	86.26
Recall (oil)	87.01	86.47
F1 (oil)	86.80	86.37
IoU (oil)	76.68	76.00
IoU (background)	89.43	89.12
mIoU	83.06	82.56
Boundary F1 (2 px)	45.52	44.65
F1 as in the official script (background class)	94.42	94.25
ROC-AUC	0.9751	0.9725
PR-AUC	0.9475	0.9455
ECE (15 bins)	0.0264	0.0324
Brier score	0.0584	0.0607

6.2 Leaky vs leak-free

Table 9. Effect of removing the leak.

Metric (%, with TTA)	Test A: official (leaky)	Test B: group split (leak-free)	Difference (pp)
Oil IoU	78.43	76.68	-1.74
Oil F1	87.91	86.80	-1.11
Precision (oil)	84.43	86.59	+2.16
Recall (oil)	91.69	87.01	-4.67
mIoU	82.79	83.06	+0.27
IoU (background)	87.15	89.43	+2.28
Accuracy	91.24	92.16	+0.91
Boundary F1 (2 px)	37.82	45.52	+7.71
ECE (×100)	2.83	2.64	-0.19
Oil share of test pixels	34.72	29.64	-5.08

- **The leak inflates the oil scores only slightly:** oil IoU -1.74 pp, oil F1 -1.11 pp. On unseen areas the model misses more oil (recall -4.67 pp) but raises fewer false alarms (precision +2.16 pp).
- **mIoU hides the drop.** The leak-free test set contains less oil (29.64 % vs 34.72 % of pixels), so background IoU rises and offsets the lower oil IoU. Oil IoU and oil F1 are the fair comparison.
- Calibration (ECE) is unchanged.

6.3 Training and qualitative results

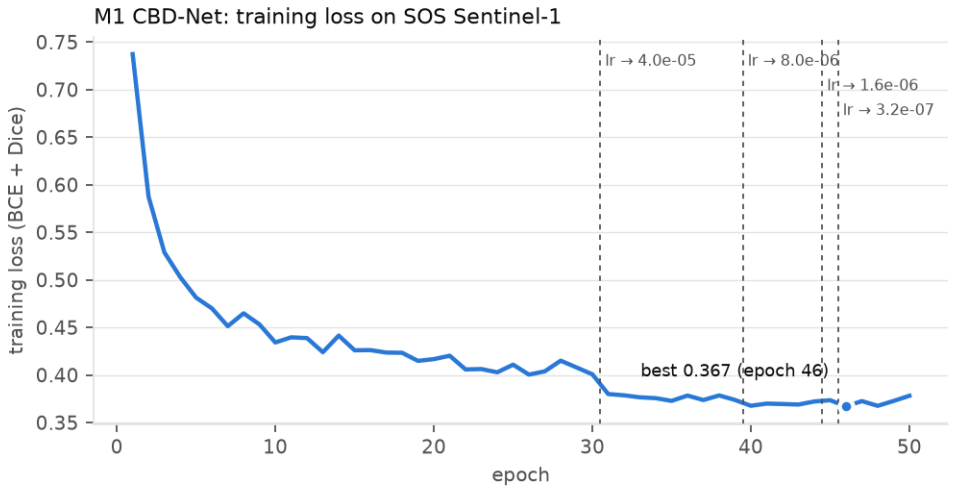


Figure 6. Training loss per epoch, leak-free split.

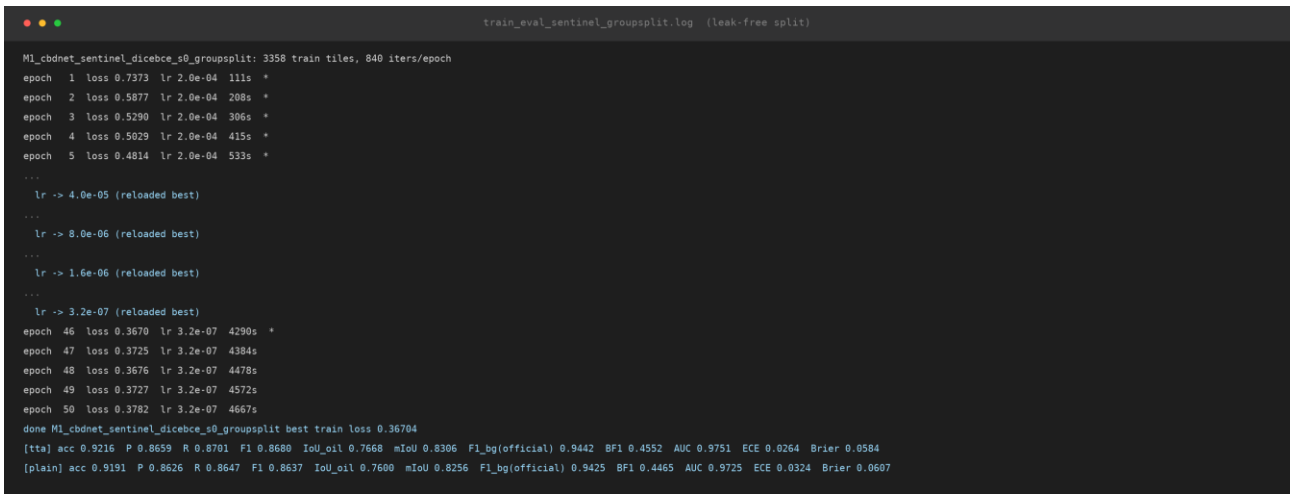


Figure 7. Console output of the leak-free run (excerpt of `train_eval_sentinel_groupsplit.log`).

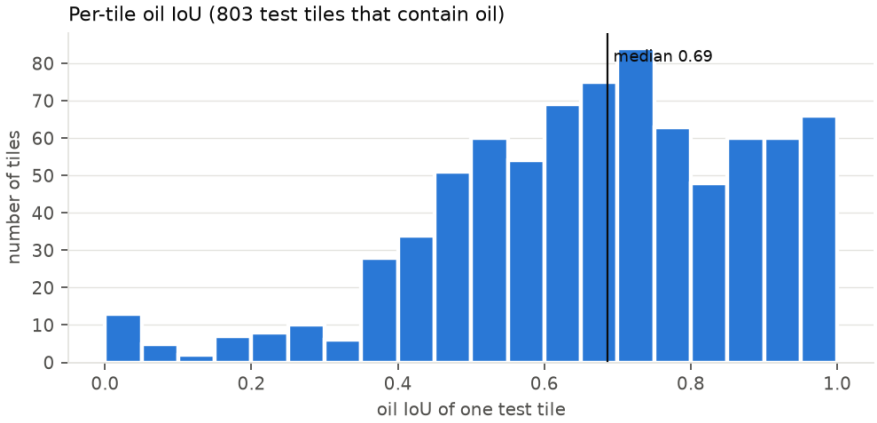


Figure 8. Per-tile oil IoU, leak-free test set. Median 0.69; 6 % of oil tiles score below 0.3.

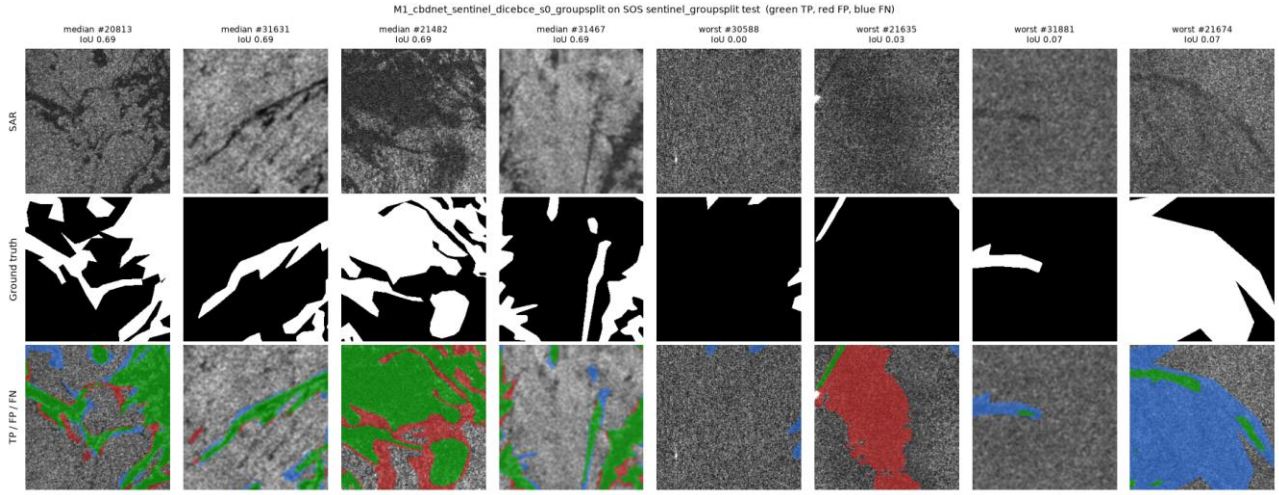


Figure 9. Four median and four worst tiles of the leak-free test set (green = correct, red = false oil, blue = missed).

- Worst tiles: (a) a large dark patch predicted as oil where only a thin strip is labelled (a possible low-wind look-alike); (b) faint thin slicks that are missed; (c) a tile labelled almost entirely as oil with no visible slick edge (label ambiguity).

Caveats. (1) One seed: differences of 1–2 pp are close to run-to-run variation, so three seeds are needed before quoting a number. (2) The groups are sea areas (crops), not the 21 original scenes, so scene-level leakage is still possible and this is a lower bound. (3) The two test sets contain different tiles, so part of any difference comes from test composition.

7. Test C: external dataset Peru (new sea)

Data. Peru oil-spill set [N22], corrected version: 2,013 Sentinel-1 patches (512 × 512) from 40 scenes never used for training. Labels: oil, look-alike, ship, land, sea. We score oil vs everything else, with the same pooled metrics and 8-way TTA as on SOS. Each patch is cut into four 256 × 256 tiles. Both models are frozen.

7.1 Quantitative results

Table 10. In-domain (SOS) vs new sea (Peru, all 2,013 patches).

Metric (%)	Official-split model: SOS	Official-split model: Peru	Leak-free model: SOS	Leak-free model: Peru
Oil F1	87.9	**24.5**	86.8	**29.4**
Oil IoU	78.4	**13.9**	76.7	**17.3**
Precision (oil)	84.4	**14.0**	86.6	**17.4**
Recall (oil)	91.7	**96.8**	87.0	**95.3**
mIoU	82.8	**37.2**	83.1	**43.5**
Accuracy	91.2	**62.8**	92.2	**71.6**
ROC-AUC (×100)	97.5	**92.1**	97.5	**92.6**
PR-AUC (×100)	95.8	**41.5**	94.7	**47.2**
ECE (×100)	2.8	**26.5**	2.6	**18.6**
Brier (×100)	6.4	**29.2**	5.8	**22.0**

Table 11. What the model calls oil on Peru: share of each true class predicted as oil (%).

True class	Share of Peru pixels	Called oil: official-split model	Called oil: leak-free model
oil	6.2	96.8	95.3
look-alike	5.3	89.9	86.0
ship	0.1	68.6	45.2
sea	84.1	37.6	27.6
land	4.3	14.3	7.7

Table 12. Peru's own scene-disjoint test subset (338 patches, 5 scenes).

Model	Oil F1	Oil IoU	Precision	Recall	ECE (×100)	Look-alike called oil
official-split	16.9	9.2	9.3	96.5	23.5	86.5
leak-free	20.0	11.1	11.2	94.9	17.7	83.5

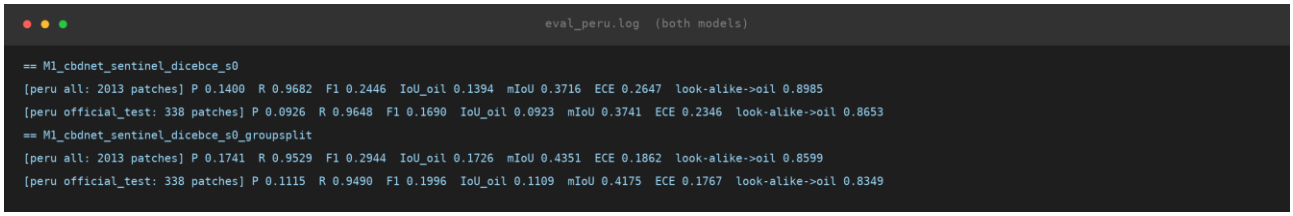


Figure 10. Console output of the Peru test, both models (`eval_peru.log`).

7.2 Qualitative results

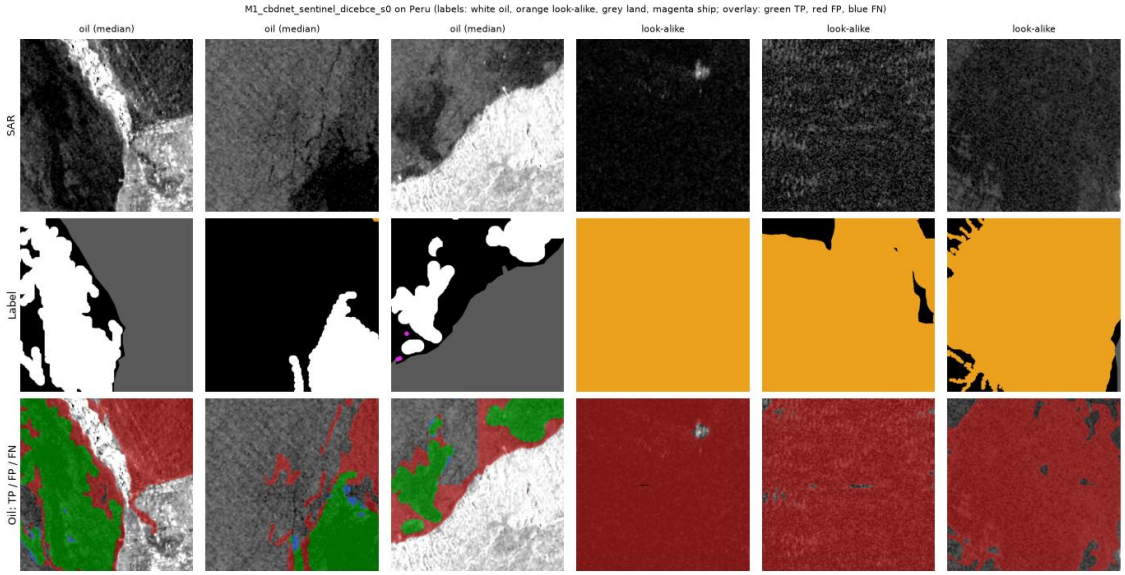


Figure 11. Peru examples, official-split model: three median oil patches and three look-alike patches. Rows: SAR, label (white = oil, orange = look-alike, grey = land, magenta = ship), prediction overlay (green = correct oil, red = false oil, blue = missed oil). The look-alike patches are predicted as oil almost everywhere.

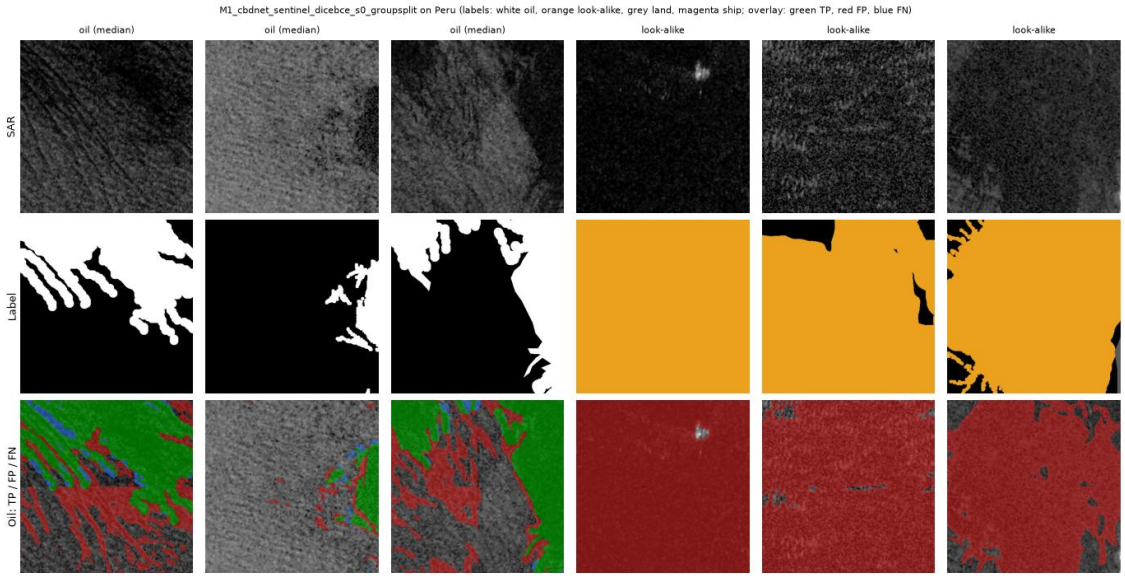


Figure 12. Peru examples, leak-free model.

7.3 Findings

- **Severe drop under domain shift:** oil F1 falls from about 87 % on SOS to 24.5–29.4 % on Peru.
- **Recall stays high but precision collapses** (14.0–17.4 %). The model calls 86.0–89.9 % of look-alike pixels oil, and about a third of open sea. Roughly 6 of every 7 oil alarms are false.
- **The ranking is still good (ROC-AUC 0.92–0.93), but the 0.5 threshold and the probabilities are wrong:** ECE rises from about 0.03 to 0.19–0.26. The model is confidently wrong on the new sea.
- **The leak-free model transfers better:** F1 +4.97 pp, ECE 0.186 vs 0.265.
- Not a data error: Peru and SOS have a similar sea brightness (median grey level 86 vs 89). Peru has lower contrast and darker oil, which is genuine domain shift.

8. Test D: external dataset MDOD (look-alike false alerts)

Data. MDOD [N05], OSF release: 4,673 full Sentinel-1 scenes with 24,940 **bounding boxes** in five classes: oil spill, biogenic slick, low wind, rain cell, sea ice. There are no pixel masks, so F1 and IoU cannot be computed. Instead we ask, for each box, how much of it the frozen model marks as oil.

Protocol. 400 boxes per class (seed 0; boxes smaller than 8 px skipped). Each box is enlarged to at least 256 × 256 for context and predicted in 256 × 256 tiles, single pass. The white land mask is excluded. **Coverage** = share of box pixels with p(oil) > 0.5. A box **alerts** if its coverage is at least 2 %. **AUC** = probability that a random oil box scores higher than a random look-alike box (0.5 = chance; below 0.5 = look-alikes score higher).

8.1 Quantitative results

Table 13. Share of boxes flagged as oil (%), per class.

Box class	Alert @2 %: official	Alert @2 %: leak-free	Alert @20 %: official	Alert @20 %: leak-free	Mean p(oil): official	Mean p(oil): leak-free
oil spill (should alert)	64.0	55.5	42.0	38.5	0.27	0.25
biogenic slick	97.0	93.2	71.2	57.0	0.36	0.27
low wind	99.8	98.2	99.5	96.8	0.64	0.56
rain cell	91.8	87.0	65.0	54.5	0.34	0.29
sea ice	96.8	91.0	83.2	66.0	0.52	0.36

Table 14. Can the model rank oil boxes above look-alike boxes? (AUC; 0.5 = chance)

Oil vs	AUC (box mean): official	AUC (box mean): leak-free	AUC (box peak): official	AUC (box peak): leak-free
biogenic slick	0.345	0.390	0.101	0.125
low wind	0.155	0.195	0.036	0.058
rain cell	0.379	0.404	0.207	0.238
sea ice	0.250	0.350	0.097	0.157
all look-alikes	0.282	0.335	0.110	0.144

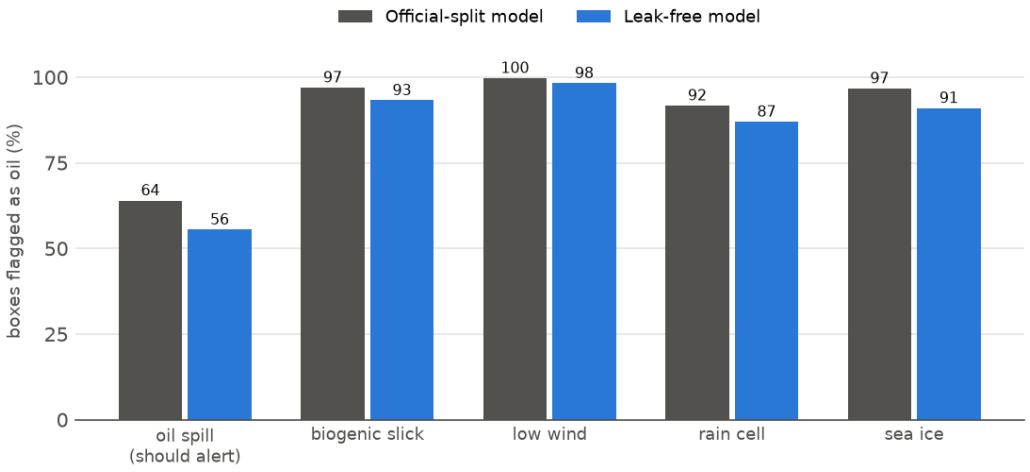


Figure 13. Share of MDOD boxes flagged as oil (coverage ≥ 2 %), per class and model.

```
evaluate_mdod.py  (summary, both models)

== M1_cbdnet_sentinel_dicebce_s0
class            n   alert@0.02  alert@0.2   mean  peak   AUC(mean) AUC(peak) oil-vs-class
biogenic slick   400   0.970      0.713      0.364 0.973  0.345     0.101
low wind         400   0.998      0.995      0.644 0.997  0.155     0.036
oil spill        400   0.640      0.420      0.273 0.639  nan       nan
rain cell        400   0.917      0.650      0.342 0.917  0.379     0.207
sea ice          400   0.968      0.833      0.521 0.968  0.250     0.097
AUC oil vs all look-alikes: mean 0.282  peak 0.11
== M1_cbdnet_sentinel_dicebce_s0_groupsplit
class            n   alert@0.02  alert@0.2   mean  peak   AUC(mean) AUC(peak) oil-vs-class
biogenic slick   400   0.932      0.570      0.273 0.947  0.390     0.125
low wind         400   0.983      0.968      0.556 0.984  0.195     0.058
oil spill        400   0.555      0.385      0.254 0.570  nan       nan
rain cell        400   0.870      0.545      0.288 0.875  0.404     0.238
sea ice          400   0.910      0.660      0.355 0.919  0.350     0.157
AUC oil vs all look-alikes: mean 0.335  peak 0.144
```

Figure 14. Console output of the MDOD test, both models (summary tables).

8.2 Qualitative results

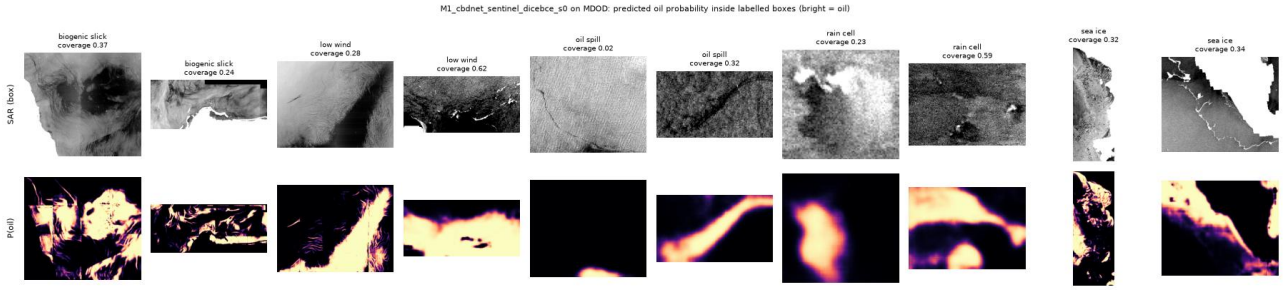


Figure 15. MDOD boxes (top) and predicted oil probability (bottom, bright = oil), official-split model.

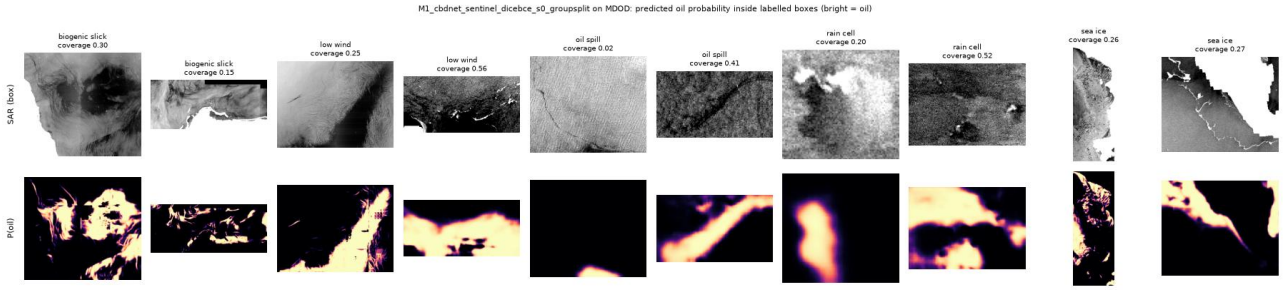


Figure 16. The same boxes, leak-free model.

8.3 Findings

- **Look-alikes trigger more alerts than real oil.** 87–100 % of look-alike boxes are flagged, but only 56–64 % of oil boxes. Low wind is the worst: almost every box is flagged, usually over more than half its area.
- **AUC is below 0.5 for every look-alike class.** M1 has learned “dark and smooth = oil”, not the shape of a slick, so large dark areas give the strongest oil signal.
- **The leak-free model raises fewer false alerts** (AUC 0.34 vs 0.28) but also flags less oil.
- **Caveat, scale:** MDOD scenes are about 10× coarser than SOS tiles, so many MDOD oil spills are thin streaks only a few pixels wide, which the model misses. Part of the low oil alert rate is this resolution mismatch. The look-alike false alerts are real at any scale.

9. Conclusions for M1

- M1 is **correctly reproduced** (mIoU and F1 within 0.2 pp of the published values).

- The leaky official SOS split overstates oil IoU and F1 by about 1–2 pp; this needs confirming with more seeds.
- In-domain scores (about 87 % F1) **do not carry over to new seas**: on Peru F1 is 24–29 %, and on MDOD look-alikes are flagged more often than oil.
- The failure mode is consistent across both external datasets: **dark areas are called oil, and the probabilities become over-confident.** The ranking (ROC-AUC 0.92 on Peru) is partly preserved, so re-calibration and better alert thresholds can recover part of the loss, while look-alike rejection needs extra information such as wind.
- The leak-free model is better on every external test and is the preferred M1 model for later comparisons.

10. Reproducing these results

All commands run from the repository root with the project virtual environment.

Table 15. Commands.

Step	Command
Test A: train + test + report	`powershell -File M1_CBDNet\run_m1.ps1`
Test B: train	`python M1_CBDNet\train_m1.py --split <group split CSV>`
Test B: test	`python common\evaluate.py --run runs\M1_..._groupsplit --split <group split CSV>`
Test C: Peru	`python common\evaluate_peru.py --run runs\<run>` then `python M1_CBDNet\make_peru_report.py`
Test D: MDOD	`python common\evaluate_mdod.py --run runs\<run>` then `python M1_CBDNet\make_mdod_report.py`
This report	`python M1_CBDNet\report\make_terminal_pics.py`, `build_m1_report.py`, then `to_pdf.ps1`

Trained weights: `runs/M1_cbdnet_sentinel_dicebce_s0/best.pth` and `runs/M1_cbdnet_sentinel_dicebce_s0_groupsplit/best.pth` (126 MB each). Leak-free split file: `leakage_check/splits/sos_sentinel_group_s0.csv`.